

Upcoming Due Dates

- **Due Today: HW10**
- **Due Thursday: Module 4 post lab worksheet**
- **Thursday will be exam 2 review day**
- **Exam 2 – Tuesday**

Why do we alloy metals?

- To increase:
 - yield and tensile strength
 - hardness
 - corrosion resistance
 - machinability, castability

**** Most metal objects are made of alloys rather than pure metals.****

Components and Phases

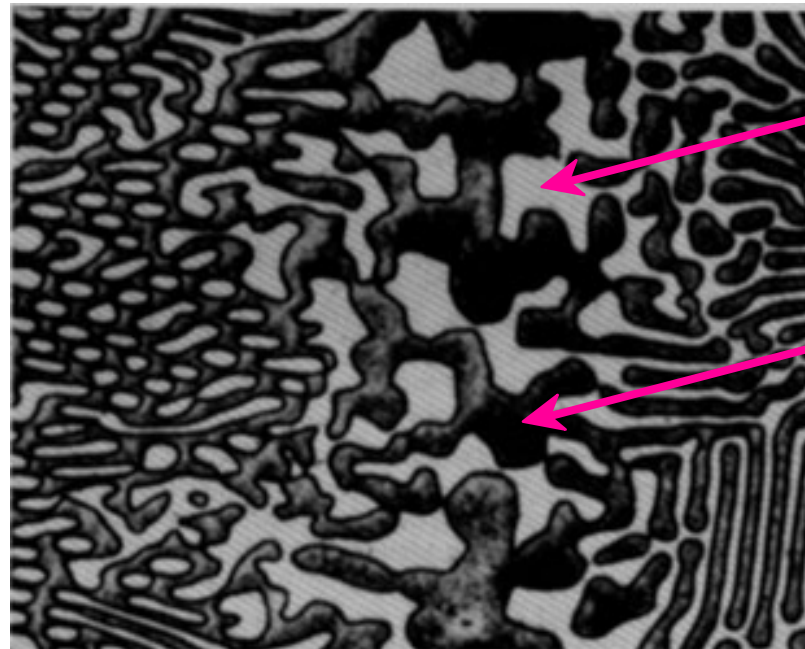
- Components:

The elements or compounds which are present in the alloy (e.g., Al and Cu)

- Phases:

The chemically homogeneous and physically distinct material regions that form (e.g., α and β).

Aluminum-Copper Alloy



β (lighter phase)

α (darker phase)

ISSUES TO ADDRESS...

When we combine two elements, what is the resulting equilibrium state?

If we specify...

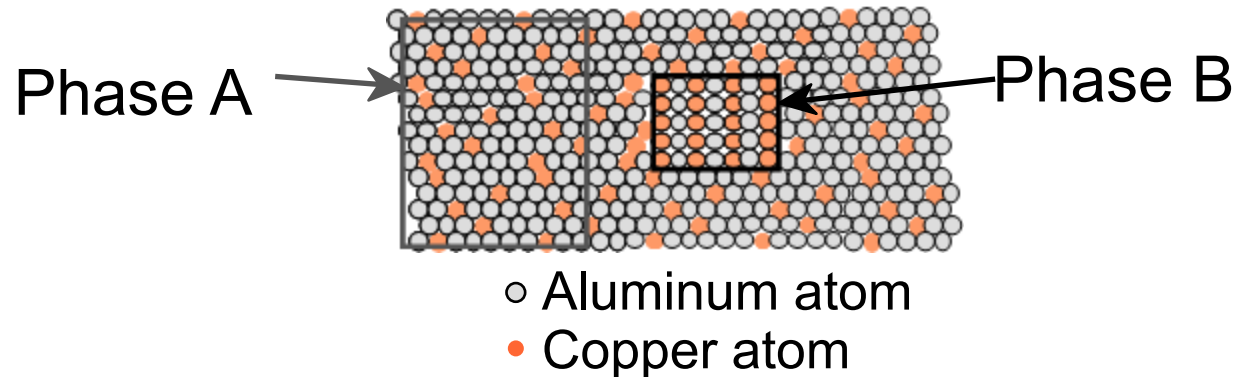
- the composition (e.g., wt% Cu - wt% Al), and
- the temperature (T)

then...

How many phases form?

What is the composition of each phase?

What is the amount of each phase?



What happens when alloying?

We need a map to show temperature, composition, and pressure during alloying

Solution: Alloy Phase Diagram

Language of phase diagrams

Component: The element or compounds in the alloy.

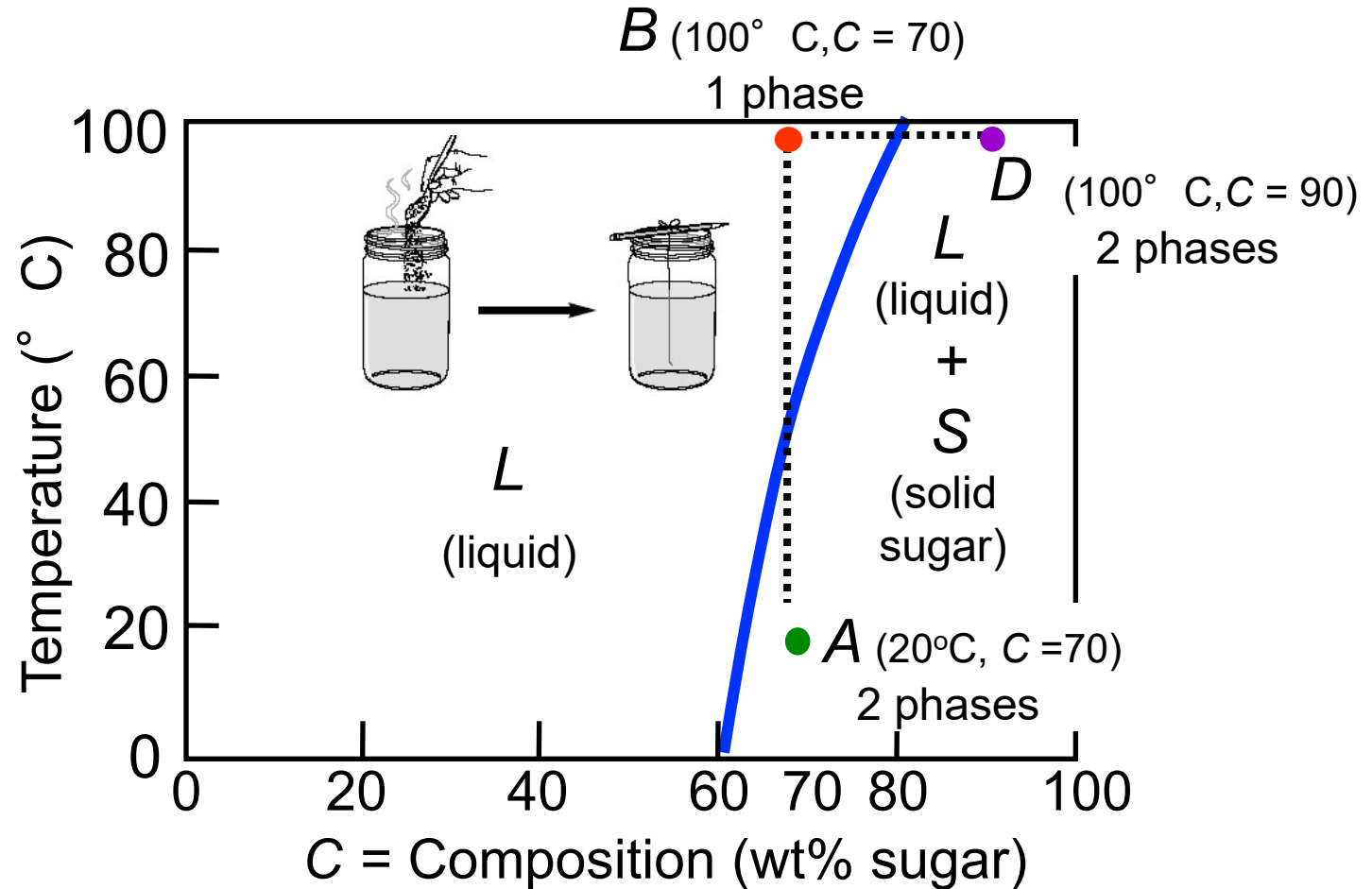
System: Components involved in a phase diagram

Phase: Chemically homogeneous and physically distinct (has a distinct and unique set of physical properties, such as crystal structure, density, strength, etc.

Altering temperature & composition can change # of phases

For a water-sugar system

- Altering T : path A to B .
- Altering C : path B to D .



Solubility Limit

- A limit that how much solute can be dissolved into a solvent (i.e. go into solution)
- For solids: atoms of the solute **substitute** on the crystal lattices of the solvent, or go into **interstices** between solvent atoms
- Greater the difference in size, and in electronegativity, the lower the solubility limit.

Criteria for Solid Solubility

Simple system (e.g., Ni-Cu solution)

	Crystal Structure	electroneg	r (nm)
Ni	FCC	1.9	0.1246
Cu	FCC	1.8	0.1278

- Both have the same crystal structure (FCC) and have similar electronegativities and atomic radii (W. Hume – Rothery rules) suggesting high mutual solubility.
- Ni and Cu are totally soluble in one another for all proportions.

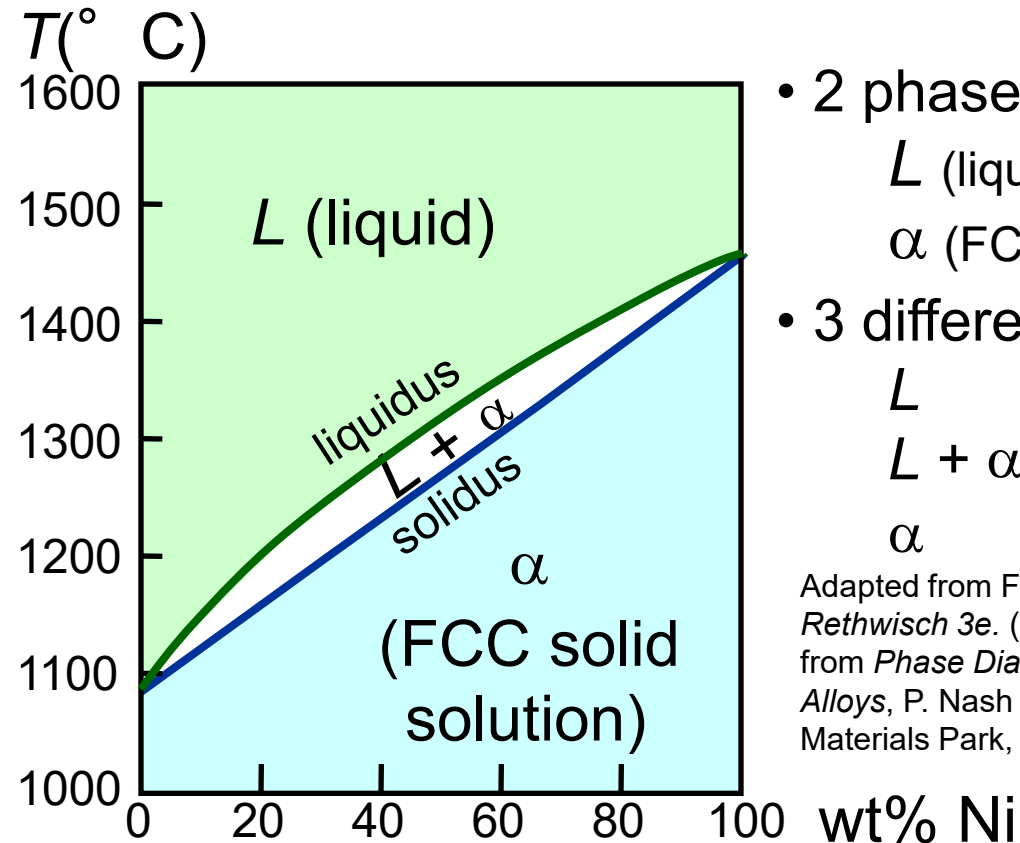
Alloying is achieved while casting



Phase Diagrams

- Indicate phases as a function of T , C , and P .
- For this course:
 - binary systems are defined as having just 2 components.
 - independent variables: T and C ($P = 1$ atm is almost always used).

Phase
Diagram
for Cu-Ni
system



- 2 phases:
 - L (liquid)
 - α (FCC solid solution)
- 3 different phase fields:
 - L
 - $L + \alpha$
 - α

Adapted from Fig. 10.3(a), *Callister & Rethwisch 3e*. (Fig. 10.3(a) is adapted from *Phase Diagrams of Binary Nickel Alloys*, P. Nash (Ed.), ASM International, Materials Park, OH (1991).

Isomorphous Binary Phase Diagram

- Phase diagram: Cu-Ni system.

- System is:

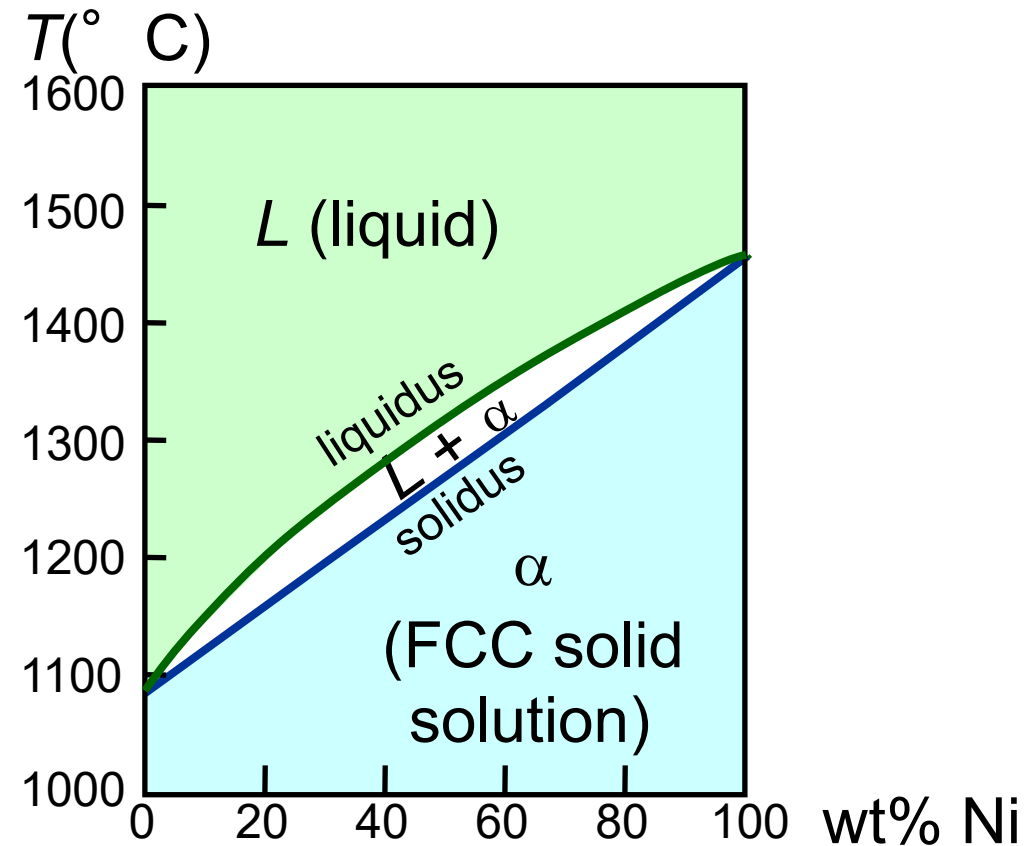
-- **binary**

i.e., 2 components: Cu and Ni.

-- **isomorphous**

i.e., complete solubility of one component in another; α phase field extends from 0 to 100 wt% Ni.

Cu-Ni phase diagram



Rule 1: If we know T and C_o , then we know: which phases are present.

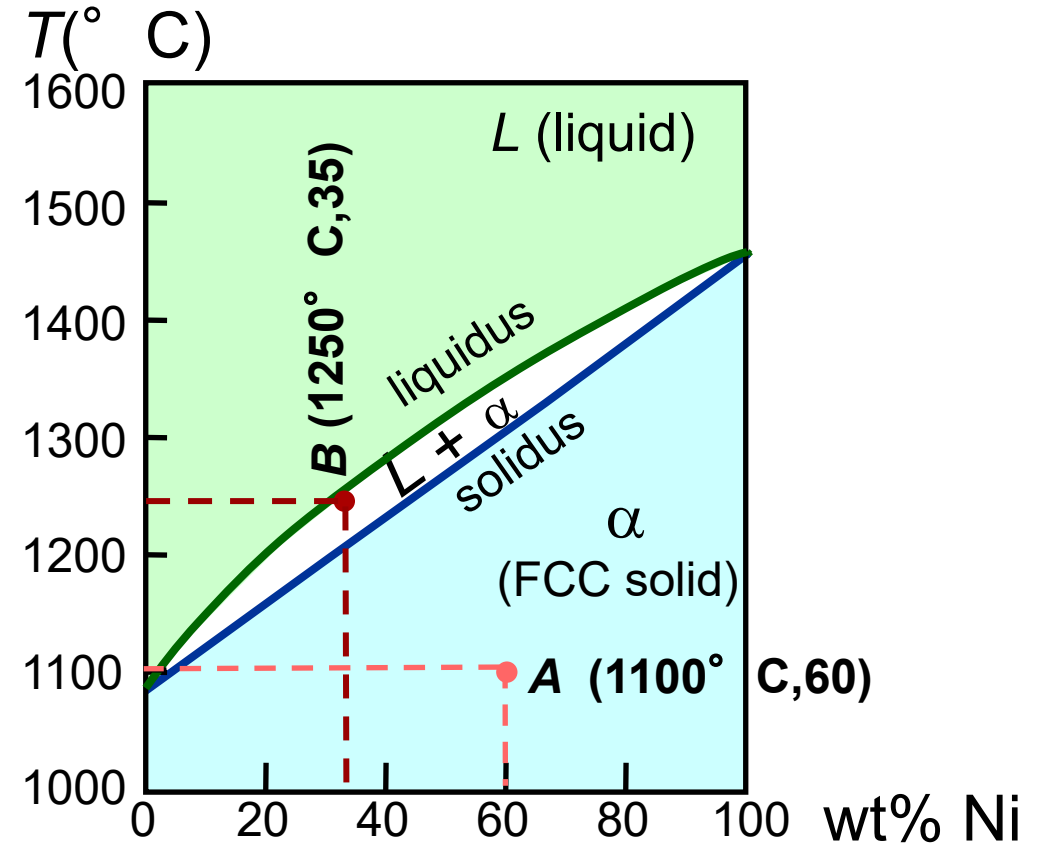
Examples:

A (1100° C, 60 wt% Ni):

1 phase: α

B (1250° C, 35 wt% Ni):

2 phases: $L + \alpha$



Rule 2: If we know T and C_0 , then we can determine the composition of each phase.

Consider $C_0 = 35 \text{ wt\% Ni}$

At $T_A = 1320^\circ \text{ C}$:

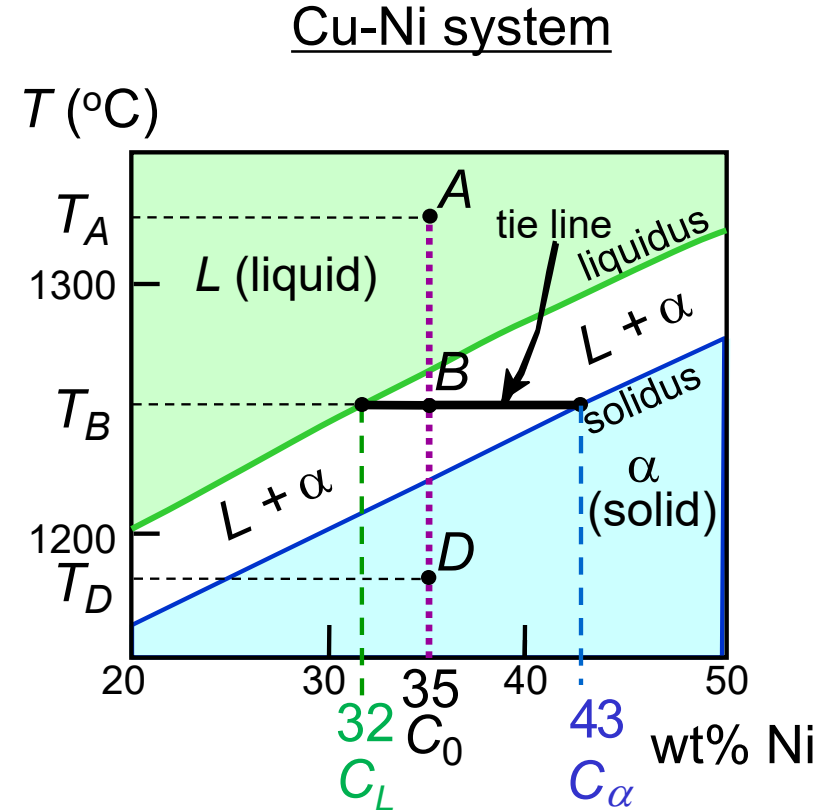
Only Liquid (L) present
 $C_L = C_0$ (= 35 wt% Ni)

At $T_D = 1190^\circ \text{ C}$:

Only Solid (α) present
 $C_\alpha = C_0$ (= 35 wt% Ni)

At $T_B = 1250^\circ \text{ C}$:

Both α and L present
 $C_L = C_{\text{liquidus}}$ (= 32 wt% Ni)
 $C_\alpha = C_{\text{solidus}}$ (= 43 wt% Ni)

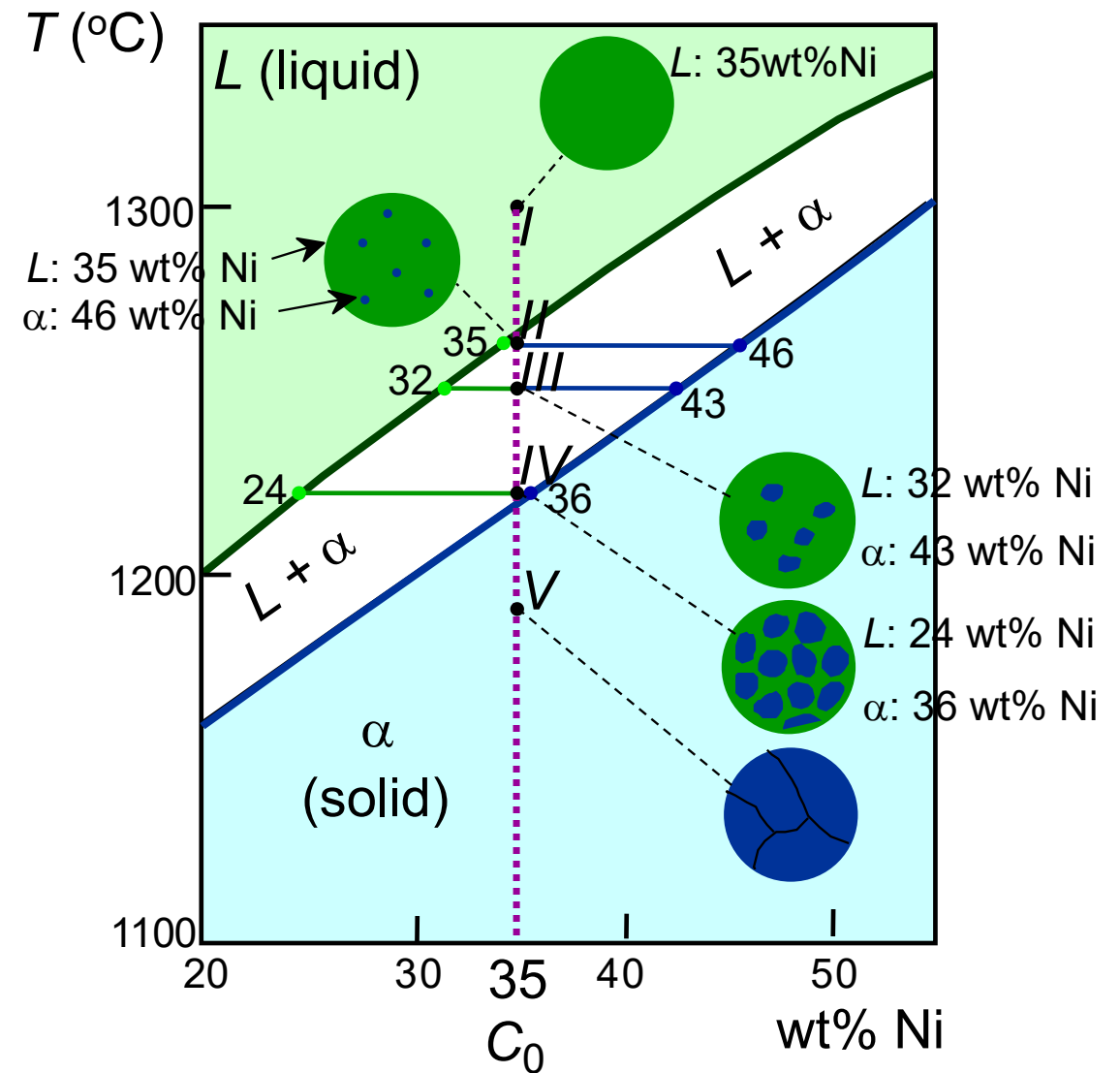


Adapted from Fig. 10.3(a), *Callister & Rethwisch 3e*. (Fig. 10.3(a) is adapted from *Phase Diagrams of Binary Nickel Alloys*, P. Nash (Ed.), ASM International, Materials Park, OH (1991).

Ex: Cooling of a Cu-Ni Alloy

Phase diagram: Cu-Ni system.

Consider microstructural changes that accompany the cooling of a $C_0 = 35 \text{ wt\% Ni}$ alloy



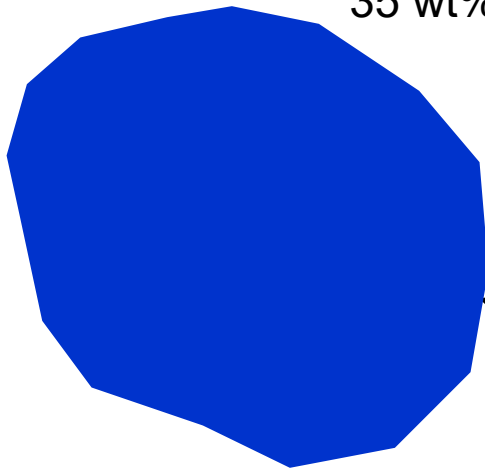
Cored vs Equilibrium Structures

C_α changes as we solidify.

Cu-Ni case: First α to solidify has $C_\alpha = 46$ wt% Ni.
Last α to solidify has $C_\alpha = 35$ wt% Ni.

Slow rate of cooling:
Equilibrium structure

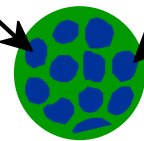
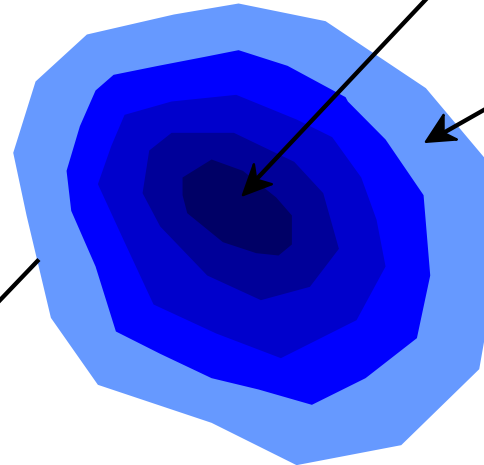
Uniform C_α :
35 wt% Ni



Fast rate of cooling:
Cored structure

First α to solidify:
46 wt% Ni

Last α to solidify:
< 35 wt% Ni

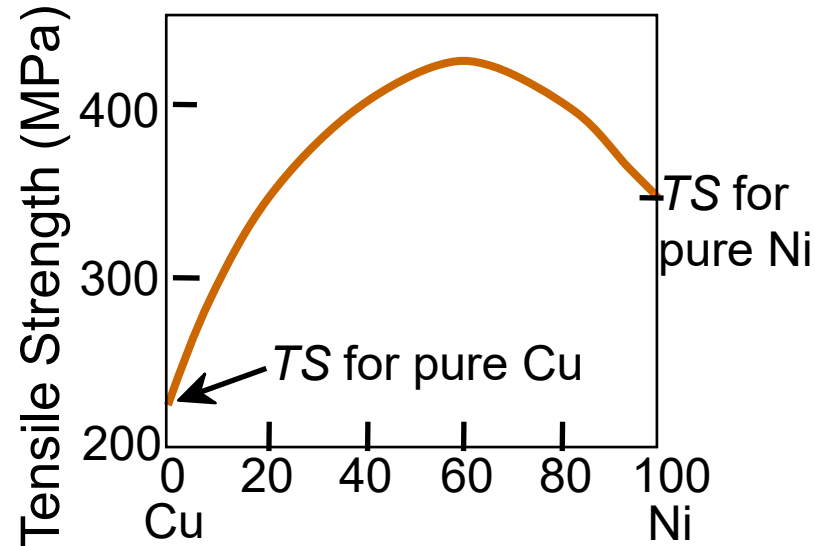


Mechanical Properties: Cu-Ni System

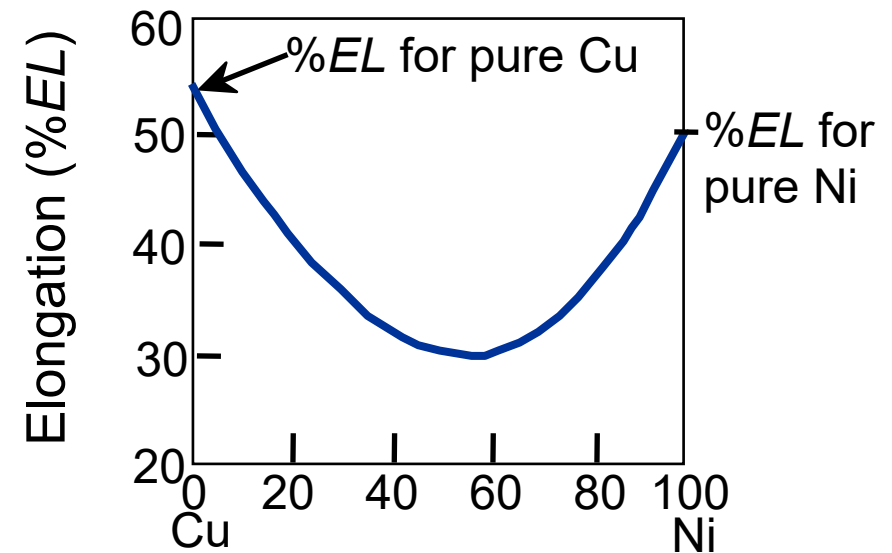
Effect of solid solution strengthening on:

-- Tensile strength (TS)

-- Ductility ($\%EL$)



Composition, wt% Ni

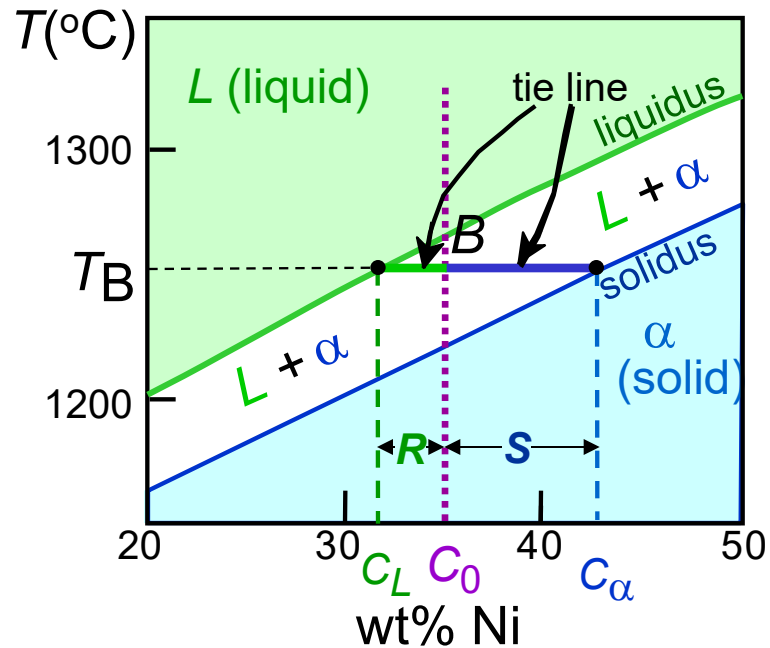


Composition, wt% Ni

The Lever Rule

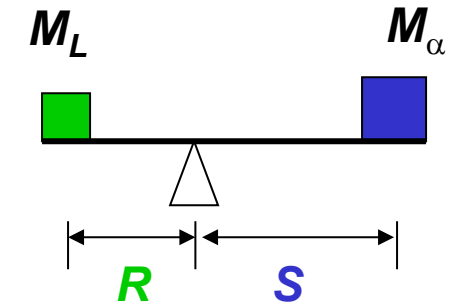
Rule 3: If we know T and C_0 , then can determine the *weight fraction* of each phase.

Tie line – connects the phases in equilibrium with each other
 – also sometimes called an **isotherm**



What fraction of each phase?

Think of the tie line as a lever (teeter-totter)



$$M_{\alpha} \times S = M_L \times R$$

$$W_L = \frac{M_L}{M_L + M_{\alpha}} = \frac{S}{R + S} = \frac{C_{\alpha} - C_0}{C_{\alpha} - C_L}$$

$$W_{\alpha} = \frac{R}{R + S} = \frac{C_0 - C_L}{C_{\alpha} - C_L}$$

Examples

Consider $C_0 = 35 \text{ wt\% Ni}$

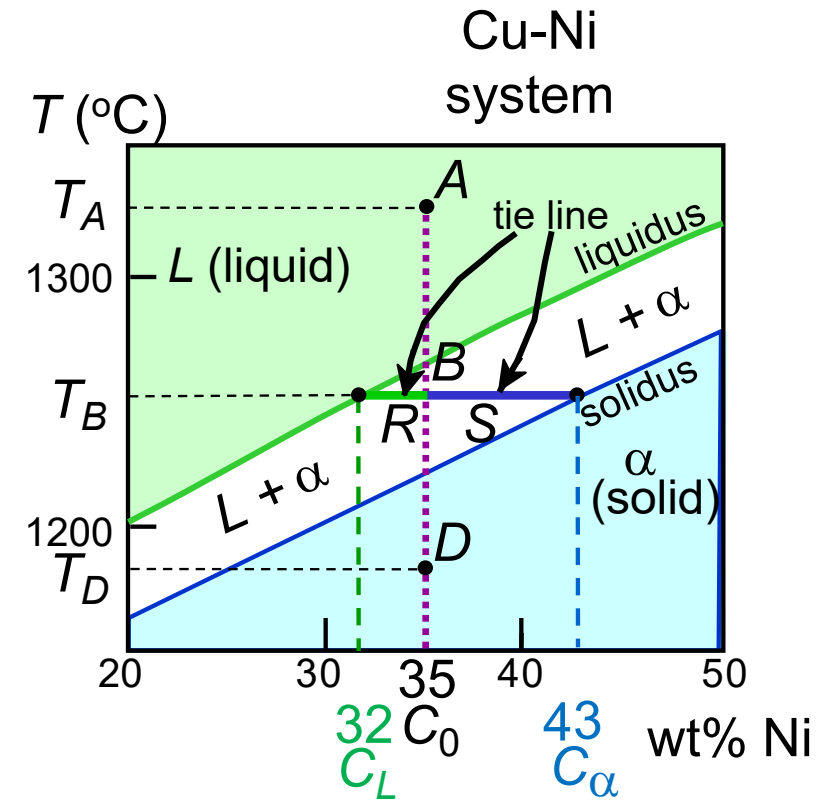
At T_A : Only Liquid (L) present

$$W_L = 1.00, W_\alpha = 0$$

At T_D : Only Solid (α) present

$$W_L = 0, W_\alpha = 1.00$$

At T_B : Both α and L present

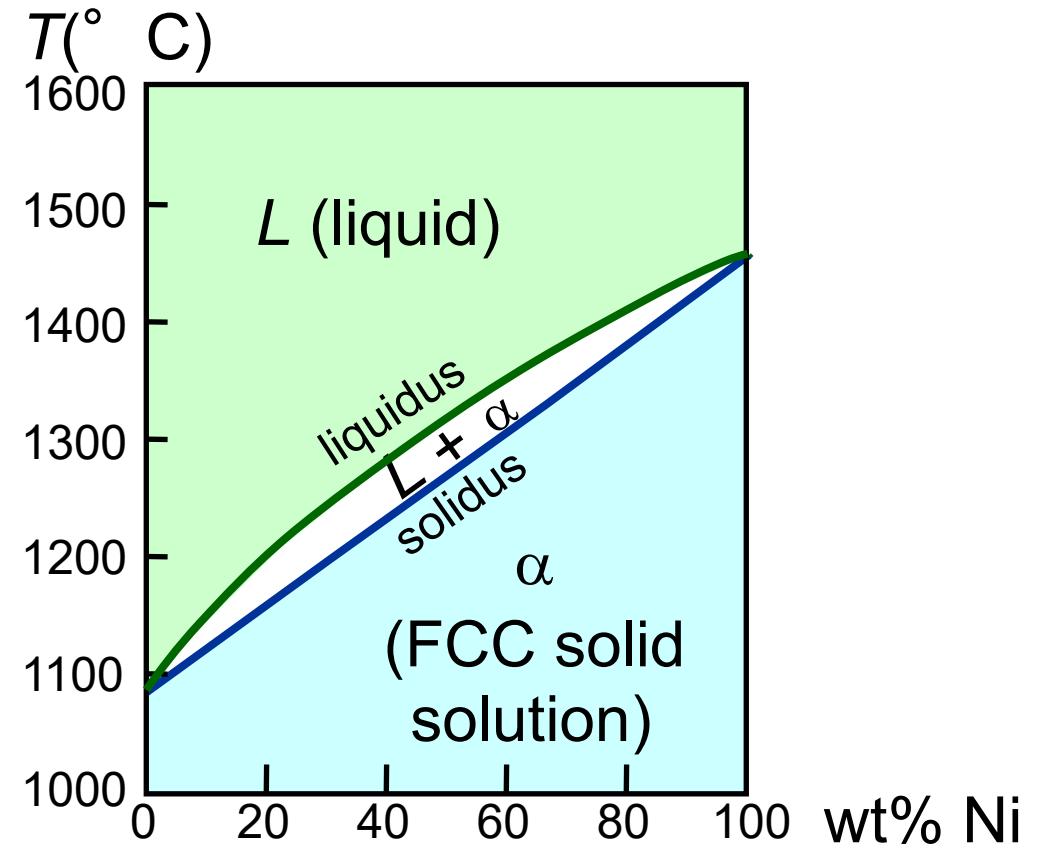


$$W_L = \frac{S}{R + S} = \frac{43 - 35}{43 - 32} = 0.73$$

$$W_\alpha = \frac{R}{R + S} = 0.27$$

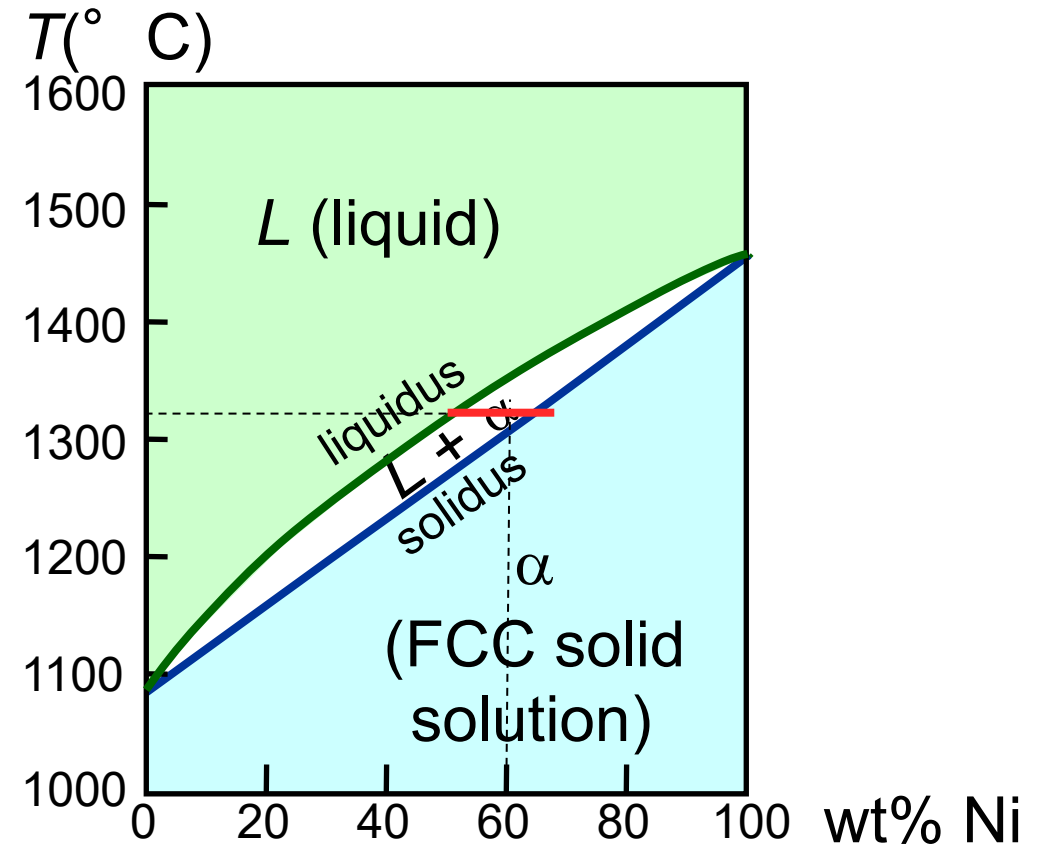
Examples

1. A Ni-Cu alloy sample composed of 60% Nickel (Ni) is slowly cooled from 1500°C to 1320°C, what phase(s) present?
2. A Ni-Cu alloy sample composed of 60% Nickel (Ni) is slowly cooled from 1500°C to 1320°C, what is the composition (%Ni) of the liquid phase and solid phase?
3. A Ni-Cu alloy sample composed of 60% Nickel (Ni) is slowly cooled from 1500°C to 1320°C, what is the mass fraction of α phase?



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